

AA-ISDN

Isosorbide Dinitrate Tablets USP 10mg Anti-anginal Agent

Pharmacodynamics: Isosorbide, a nitrate relaxes vascular smooth muscle. The efficacy of nitrates in alleviating the pain of angina pectoris is probably due to a reduction in myocardial oxygen demand rather than to an increase in myocardial oxygen supply. Although venous effects predominate, nitrates dilate both arterial and venous beds. Dilating the post-capillary vessels including large veins, promotes a peripheral pooling of blood and decreases venous return to the heart. These reduces left ventricular end-diastolic pressure (pre-load). The arterial pressure (after-load) and decrease in ventricular volume reduces intramyocardial tension and lessens myocardial oxygen demand.

Pharmacokinetics: After a single oral dose of 30mg isosorbide mean plasma peak concentrations of 24ng/ml were observed within 30 minutes.

Indications: For the prophylaxis of ischemic heart pain associated with chronic angina of effort. It may reduce the number, duration and severity of anginal attacks increase exercise tolerance and curtail nitroglycerin requirements.

Contraindications: Hypersensitivity to isosorbide or known idiosyncratic reaction to organic nitrate. Increased intracranial pressure (e.g. head trauma or cerebral hemorrhage). Withhold treatment in the presence of cardiogenic shock or if there is a risk of shock developing.

Route of administration: Oral

Drug Interactions: The effectiveness of sublingual isosorbide dinitrate was reduced in patients taking disopyramide, tricyclic antidepressants and antimuscarinics such as atropine and propantheline owing to diminished salivary secretion.

Warnings: The data supporting the use of nitrates during the early days of the acute phase of myocardial infarction (the period during which clinical and laboratory findings are unstable) are not sufficient to establish safety. Nitrate dependence may occur with chronic use. To avoid possible withdrawal effects, gradually reduce administration. Do not discontinue abruptly. Chest pain, acute myocardial infarction and even sudden death have occurred during a temporary withdrawal of nitrate exposure in industry workers continuously exposed to nitrates.

Precautions: Tolerance to isosorbide and cross tolerance to other nitrates and nitrites may occur. Administer isosorbide with caution to patients with glaucoma. Isosorbide like nitroglycerin is a potent vasodilator and causes a slight decrease in mean blood pressure (approximately 10 to 15mm/Hg) in some patients when used in therapeutic dosages. Exercise caution using the drug for patients who are prone to or might be affected by hypotension.

Usage during pregnancy and lactation is not known.

Adverse Effects: Headache is the most frequent adverse effect and it may be severe and persistent. This occurs most frequently at the beginning of therapy. Usually a temporary reduction in dosage will control headache along with a concomitant administration of commonly used analgesics or by administering the drug during meals. These headaches disappear within a week of continuous uninterrupted therapy. Advise the patient that they may occur and that isosorbide is important with regard to preventing of angina. Occasionally drug and/or exfoliative dermatitis occur. Signs of cerebral ischemia associated

with postural hypotension such as weakness and transient episodes of dizziness, may occasionally develop. Cutaneous vasodilation with flushing may occur. Rarely, a marked sensitivity to the hypotension effects of the drug can occur with a severe response (nausea, vomiting, restlessness, perspiration and collapse) and alcohol may enhance this effect. Isosorbide can antagonize the effects of histamine, acetylcholine and similar agents.

Overdose: Symptoms: Related to vasodilation: cutaneous flushing, headache, nausea, dizziness and hypotension. Treatment: Gastric lavage with symptomatic and supportive therapy, including ventilation with oxygen and vasopressor amines if indicated.

Dosage: Individualize dosage as there are wide inter-patient differences in sensitivity to isosorbide dinitrate. Usual dosage range is 5mg to 30mg 4 times daily titrated according to therapeutic and patient response.

Supplied:

AA-ISDN Tablets 10mg

White, round tablet, bevelled-edge & flat one side, convex other. Engraved score over 10 on flat side, convex side plain.

Available in HDPE bottles of 100 tablets.

Storage Conditions:

Store below 30°C

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